

Dhruv Rath

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SUMMARY

Final-year B.Tech student building AI/ML applications across Generative AI, Agentic AI, and backend AI systems. Experienced in developing LLM-powered systems, RAG pipelines, predictive models, and full-stack applications. Proficient in Python, Java, and SQL.

EDUCATION

Vellore Institute of Technology, Chennai

2023 – present (2027)

B.Tech - Electronics and Computer

CGPA - 9.06

Relevant Courses: Data Structures & Algorithm, OOP, Database Management, Operating Systems, Computer Networks, Computer Architecture, Cloud Computing, AI & ML, Deep Learning, Web Programming

EXPERIENCE

AI & ML Intern

June 2026 – August 2026

Sutherland

Chennai, Tamil Nadu

- Engineered FastAPI services and data pipelines for LLM-powered retrieval and forecasting workflows.
- Developed Agentic AI workflows with modular tools and task orchestration for LLM automation.
- Architected a contractual-document RAG pipeline with hybrid retrieval, cross-encoder reranking, and citation-aware LLM generation, prototyped on 100 documents and designed to scale to 1M documents.

SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript

AI & GenAI: RAG, Agentic AI, Vector Embeddings, Feature Engineering, Exploratory Data Analysis (EDA), LLMs, Prompt Engineering, NLP, Machine Learning, Deep Learning

Frameworks & Tools: LangChain, FastAPI, ChromaDB, PyTorch, TensorFlow, Keras, Scikit-learn, Streamlit, NumPy, Pandas, MySQL, Git, GitHub

PROJECTS

DocMate – Multi-Format Document Assistant [GitHub](#) *Python, FastAPI, LangChain, ChromaDB, React*

- Architected a full-stack RAG assistant supporting multi-format parsing for high-fidelity, context-aware Q&A.
- Built a hybrid retrieval and reranking pipeline pairing ChromaDB dense vectors with BM25 keyword search, refined by a cross-encoder reranker and confidence thresholds to eliminate hallucinations.
- Implemented session-isolated vector pipelines enabling scalable, multi-user document intelligence.

TripPilot – Agentic Trip Planner [GitHub](#) *Python, Groq LLM, Pydantic, SerpApi, RouteStack, Open-Meteo*

- Built an LLM-powered agentic workflow for constraint-aware trip planning using live travel APIs.
- Orchestrated concurrent API calls with structured data processing, candidate ranking, and budget-constrained selection.
- Implemented adaptive replanning and deterministic validation for reliable itinerary generation.

PayCast – Invoice Payment Forecasting [GitHub](#) | [Live](#) *Python, Scikit-learn, Streamlit, Numpy, Pandas*

- Designed an analytics system to forecast invoice clearing timelines across 50,000+ business transactions.
- Built an automated preprocessing and model selection pipeline with recursive feature elimination.
- Compared Random Forest, Gradient Boosting, and XGBoost, achieving R^2 0.656 and RMSE 8 days.

CERTIFICATIONS

- IBM – Advanced Generative AI - Professional
- Oracle – Agentic AI Certified Foundations Associate
- IBM – Prompt Engineering for Everyone
- Infosys Springboard – Machine Learning Implementation

ACHIEVEMENTS

- Shortlisted among 416 of 3,01,066 teams (**Top 0.14%**) in the Adobe University Hackathon 2026.
- Design Lead for the Sports Club at VIT Chennai